SCHOOL OF MATHEMATICS

First Assessment 2079

Subject: **Techniques for Big Data**

Course No: **MDS 603**

Level: **MDS /II Year /III Semester**

Full Marks: 45

Pass Marks: 22.5

Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [3 ×5 = 15]

1. What do you mean by Big Data? How big the data would you consider as Big Data?
2. Briefly explain the different characteristics of Big Data.
3. What is functional Programming? What are the concepts borrowed by Map Reduce from Functional Programming?
4. When mapper worker is failed both in-progress and completed tasks are reset to idle but when Reducer worker is failed only in-progress tasks are reset to idle. What is the reason behind this?
5. What is Hadoop? Briefly explain about the history behind the development of Hadoop.

Group B [5x6=30]

6. What are the scopes and challenges of Big Data? Explain with suitable example.

OR

Explain about the current trend of Big Data along with its applications in the Covid 19 pandemic.

7. Explain the Distributed Execution Overview of Map Reduce.
8. Draw the block diagram to find word frequency in following text:

   Big Data Technology uses Big Data Analytics

   Big Volume of Data Gives Big Value

   Big Data has Velocity Variety and Veracity

   Big Data is really Big Technology

9. What is Hadoop Ecosystem? Explain about core components of the Hadoop Ecosystem.
10. Explain the Master/Slave Architecture in Hadoop

OR

Explain the different configuration modes to setup the Hadoop.

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Second assessment 2079

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Full Marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answer in their own words as far as practicable.

Attempt ALL questions

Group A [5x3=15]

1. What is NoSQL? How is it different than SQL?
2. Compare and Contrast HBase with HDFS.
3. What is Spark? How is it different than MapReduce?
4. Compare Pig Latin with SQL.
5. Differentiate between Hive QL and SQL

Group B [5x6=30]

6. Explain the CAP Theorem with example.

OR

What are the different types of NoSQL Database? Explain with example.

7. Can Mongo DB be considered as an alternate of RDBMS? Compare it against SQL with example.
8. How could SQL queries be executed in Spark. Explain with example.
9. What are the different execution modes in Apache Pig? Explain.
10. Explain about Hive Shell, Hive Services and Hive Metastore.

OR

Explain the user defined functions in Apache Pig and Apache Hive.

First Assessment 2080

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5x3=15]**

1. Explain about the different characteristics of big data.
2. How the functional programming language of MapReduce different from the object-oriented programming language?
3. What are the core components of Hadoop?
4. What do you understand by the term "running Hadoop"?
5. Explain about different map reduce input and output formats.

Group B **[5x6=30]**

6. What is map reduce? Explain its distributed execution overview.
7. Explain about the anatomy of read and write operation ofthe Hadoop Distributed File System (HDFS).

OR

Explain about the HDFS architecture with its running daemons.

8. Explain about the anatomy of map reduce job run, failures, shuffle and sort with reference to word frequency count example.

OR

What are the different configuration modes to setup the Hadoop? Which setup mode is preferred for the development and why? Also briefly explain about Hadoop streaming.

9. Write mongodb query for the following database:

➢ db.order.find({})

```
[{
cust_id: "ID1",  "ord_date": ISODate("2018-05-04"),  "price": 400,  "status": "A",  "itemqty": 20
},
{
cust_id: "ID1",  "ord_date": ISODate("2020-05-04"),  "price": 400,  "status": "not A",  "itemqty": 20
},
{
cust_id: "ID2",  "ord_date": ISODate("2015-05-04"),  "price": 200,  "status": "not A",  "itemqty": 10
}]
```

a) For each unique cust_id, sum the price field on your created order collection
b) For each unique cust_id, ord_date grouping, sum the price field. Exclude the time portion of the date.
c) For cust_id with multiple records, return the cust_id and the corresponding record count.
d) For each unique cust_id, ord_date grouping, sum the price field and return only where the sum is greater than 250. Exclude the time portion of the date.
e) For each unique cust_id with status A, sum the price field
f) For each unique cust_id with status A, sum the price field and return only where the sum is greater than 250.

10. Give some examples of HDFS commands. What is the single point of failure in Hadoop version 1 and how Hadoop version 2 tries to solve it? Explain.

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Second Assessment 2081

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5x3=15]**

1. What is NoSQL? How is it different than SQL?
2. How Apache Spark is different from Hadoop Mapreduce?
3. Explain different execution modes of Apache Pig.
4. Explain about HBase architecture briefly.
5. What is the process to create UDFs (User Defined Functions) in Apache Pig?

Group B **[5x6=30]**

6. Explain about the CAP theorem. What is mean by Eventual 6. 6. Consistency in NoSQL databases.

OR

What are the different types of NoSQL databases? Explain with examples.

7. Write short notes on:
   a) Spark SQL
   b) Spark MLib
8. Explain architecture of Apache Spark. Also explain about spark transformations and actions with examples.
9. Explain about Hive Shell, Hive Services and Hive Metastore.

OR

Explain about Hive architecture in detail. Also, compare Apache Hive with Apache Pig.

10. Write mongodb query for the following database:
    ➢ db.order.find({})

```
[
{ "customer": "Alice", "product": "Laptop", "price": 1200, "date": "2024-03-01" },
{ "customer": "Bob", "product": "Phone", "price": 800, "date": "2024-03-01" },
{ "customer": "Alice", "product": "Tablet", "price": 500, "date": "2024-03-02" },
{ "customer": "Charlie", "product": "Laptop", "price": 1000, "date": "2024-03-02" },
{ "customer": "Bob", "product": "Laptop", "price": 1300, "date": "2024-03-03" },
{ "customer": "Alice", "product": "Phone", "price": 700, "date": "2024-03-03" },
{ "customer": "Bob", "product": "Tablet", "price": 600, "date": "2024-03-03" }
]
```

   a) Write a query to find maximum price per customer.
   b) Write a query to find total revenue per day.
   c) Write a query to find average price per customer per product.

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2081

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Master Level / Second Year /Third Semesters
Data Science (MDS 603)
(Techniques for Big Data)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as for as practicable.
Attempt All Questions

Group A **[5 × 3 = 15]**

1. Explain about the different characteristics of big data.
2. How the functional programming of MapReduce different from imperative programming?
3. What are the core components of Hadoop?
4. Explain about the different types of NoSQL databases.
5. Explain about the different configuration modes of Apache Pig.

Group B **[5×6 = 30]**

6. Explain about the anatomy of a MapReduce job run, failures, shuffle and sort with reference to word frequency count example.

7. Explain about the anatomy of write operation ofthe Hadoop Distributed File System (HDFS).

OR

Explain about the HDFS architecture and HDFS commands.

8. Explain about the CAP theorem. What do you mean by "Eventual Consistency" in NoSQL databases?

OR

Write short notes on:

a) Resilient Distributed Datasets
b) Spark Streaming

9. Write mongodb query for the following database:

> ➢ db.order.find({})

[{ cust_id: "ID1", "ord_date": ISODate("2018-05-04"), "price": 400, "status": "A", "item qty": 20},
{ cust_id: "ID1", "ord_date": ISODate("2020-05-04"), "price": 400, "status": "not A", "item qty": 20},
{ cust_id: "ID2", "ord_date": ISODate("2015-05-04"), "price": 200, "status": "not A", "item qty": 10},
...]

a) For each unique cust_id, sum the price field and sort it by sum.
b) For each unique cust_id, and unique day, month and year, sum the price field.
c) For each unique cust_id with status A, sum the price field and return only those records where the sum is either 250 or 300.

10. Explain about the architecture of Apache Hive. Also, compare Apache Hive with Apache Spark and Apache Pig.

Institute ...

2079

Master Level / Second Year /Third Semester/ Science
Data Science (MDS 603)
(Techniques for Big Data)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as for as practicable.

Attempt All Questions

Group A **[5 × 3 = 15]**

1. What are the different characteristics of big data?
2. How is functional programming language different than imperative programming language?
3. What are the core components of Hadoop Ecosystem?
4. What are structured, unstructured and semi structured data?
5. How Spark is different than Map Reduce? Compare Hive QL with SQL.

Group B **[5×6 = 30]**

6. Explain the scopes and applications of big data analytics in the different sectors with example.
7. What is map reduce? Explain its execution overview with reference to word frequency count example.
8. What are the different daemons running in a Hadoop cluster? Explain how these daemons work in Master/Slave Architecture.

OR

What are the different configuration modes to setup the Hadoop? Which setup mode is preferred for development and why?

9. Explain the limited taxonomy of NoSQL data base with example.

OR

What are the limitations of distributed databases like NoSQL? Explain it with reference to CAP Theorem.

10. Explain about resilient distributed dataset and data frames in spark.

Institute of Science and Technology
2082

MDS / II Year / Third Semester/ Science
Data Science (MDS 603)
(Techniques for Big Data)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks. The symbols have their usual meanings.
Attempt all questions.

Group A [5 × 3 = 15]

1. What do you understand by big data? What are the common misconceptions about big data?
2. List out the key features of Apache Hadoop.
3. Differentiate between HBase and HDFS.
4. What is Spark Shell used for? What are the components of Spark Shell?
5. What are the use cases of Apache Pig?

Group B [5 × 6 = 30]

6. What are Hadoop Daemons? Explain the various Hadoop Daemons.
7. Explain HDFS replica placement strategy and how rack-awareness improves fault tolerance.
8. Explain the architecture of HBase along with its data model.

OR

Differentiate between NoSQL database and Relational Database.

9. What do you mean RDD? When should you use RDD? Explain the operations involved in RDD.
10. With an appropriate diagram, explain the components of Hive's architecture and their functions.

OR

Explain execution modes Apache Pig. Differentiate Pig with SQL database.

First Assessment 2081

Subject: **Technique for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5 × 3 = 15]**

1. What do you mean by big data? Briefly point out the challenges of implementing big data.
2. What is Hadoop Streaming? What is it used for? 2.5
3. Write the key features of Apache Hadoop. 2.5
4. How does HDFS works? 2
5. What are the components of Spark Shell? What are they used for? 2 11

Group B **[5 × 6 = 30]**

6. Explain the characteristics of big data. 5
7. Explain the operations of RDD. Distinguish between RDD, DataFrame and Dataset.

OR

What is Structured Streaming? Explain the working mechanism of Structured Streaming?

8. Explain the various configuration of Hadoop along with their characteristics. 5
9. Explain the replica placement and risk-aware policy in HDFS.

OR

Explain the architecture of HDFS with block diagram. 5

10. Explain the key components of Hadoop. 5

Second Assessment 2082

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
PassMarks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What are the defining characteristics of big data?
2. When is it appropriate to use the Hadoop Federation configuration?
3. What types of failures does MapReduce address?
4. What is the function of ZooKeeper in HBase?
5. What are the main features of RDD Transformations?

Group B [5x6=30]

6. What does big data mean? Discuss the current trends and challenges surrounding big data.
7. Explain Apache Hadoop and its main components.
8. What is MapReduce? Describe the process of executing a MapReduce job on a cluster.

OR

Describe HDFS, its working process, and its architecture.

9. Compare and contrast NoSQL and relational databases.

OR

Explain the read and write operations in HBase.

10. Describe the architecture of Hive and its key components.

First Assessment 2082

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5x3=15]**

1. Explain about the different characteristics of big data.
2. What is the usage of map() and reduce() function in mapreduce?
3. What are the core components of Hadoop?
4. What is the single point of failure in Hadoop version 1 and how Hadoop version 2 tries to solve it? Explain.
5. Explain about different map reduce input and output formats.

Group B **[5x6=30]**

6. Explain about distributed execution overview of mapreduce.
7. Explain about the anatomy of read and write operation of the Hadoop Distributed File System (HDFS).

OR

Explain about the HDFS architecture with its running daemons.

8. Explain about the anatomy of map reduce job run, failures, shuffle and sort with reference to word frequency count example.

OR

What are the different configuration modes to set up Hadoop? Which setup mode is preferred for the development and why? Also briefly explain about Hadoop streaming.

java to hadoop code

9. Explain the Hadoop ecosystem. Why was the Zookeeper introduced? ?
10. Write twelve HDFS commands with examples.

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Second Assessment 2083

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full Marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What is a Resilient Distributed Dataset (RDD) in Apache Spark?
2. Explain three key characteristics that make Spark's performance superior to Hadoop Map Reduce.
3. Explain the different execution modes available in Apache Pig.
4. Differentiate between HBase and HDFS.
5. Write short notes on Spark Streaming with examples.

Group B [5x6=30]

6. Write all mongodb queries for each of these questions (each question carries 1 mark):
   Collection Name: employees
   - name (String) - Full name of the employee.
   - dept (String) - Department name (e.g., "IT", "HR").
   - hireYear (Number) - Year they joined (e.g., 2019).
   - salary (Number) - Annual salary in dollars.
   - skills (Array of Strings) - List of technical skills (e.g., ["Java", "Python"]).
   - projects (Nested Object) - Contains assigned (Number) and completed (Number). e.g. projects: {"assigned": 12, "completed": 2}
   - resigned (Boolean) - Optional field. Indicates if they left the company.

   a) Find employees hired exactly in the year 2020. Show only the name and dept fields, and hide the id field.
   b) Find employees who have exactly both "Java" and "Python" included in their skills array.
   c) Find all employees whose name starts with the word "John".
   d) Find all employees where the number of projects.completed is exactly equal to the number of projects.assigned.
   e) Find the average salary of employees specifically working in the "IT" department.
   f) Flatten the skills array to find the most popular skill in the company. Count the total occurrences of each skill and sort them from highest to lowest.
7. Explain spark architecture in detail with a diagram. Why Apache Spark uses lazy evaluation strategies to execute programs.

   OR

   Explain about the ecosystem or unified stack of Apache Spark in detail. What is DAG and how it is executed in spark.
8. Explain architecture of HBase in detail with its single point of failure. Is NoSQL a direct replacement for SQL? Provide a technical justification for your stance.

   OR

   Explain fundamental differences between HBase and a traditional Relational Database Management System (RDBMS) in terms of schema, transactions, and data volume handling. Also write queries to create your own table, insert data and retrieve data to and from HBase.
9. Explain Brewer's theorem with an example.
10. Explain Hive architecture in detail with Hive services and Hive metastore. Differentiate between Apache Pig and Hive.

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Second Reassessment 2082

Subject: **Techniques for Big Data**
Course No.: **MDS 603**
Level: **MDS /II Year / III Semester**

Full Marks: 45
Pass Marks: 18
s Time: 2 hrs

Candidates are required to give their answers in their own word as far as practicable
Attempt ALL questions.

Group A [5 x 3 =15]

1. Explain the importance of RDD in Apache Spark? Marks: 3
2. Differentiate between Apache Spark and Hadoop Map-reduce. Marks: 3
3. Explain the architecture of Apache Pig with a diagram. Marks: 3
4. Explain transformations and actions in Apache Spark with examples. Marks: 3
5. Write short notes on Spark SQL with examples. Marks: 3

Group B [5 x 6 =30]

6. Write all mongodb queries for each of these questions (each question carries 1 mark): Marks: 6
   Collection Name: employees
   - name (String) - Full name of the employee.
   - dept (String) - Department name (e.g., "IT", "HR").
   - hireYear (Number) - Year they joined (e.g., 2019).
   - salary (Number) - Annual salary in dollars.
   - skills (Array of Strings) - List of technical skills (e.g., ["Java", "Python"]).
   - projects (Nested Object) - Contains assigned (Number) and completed (Number). e.g. projects: {"assigned": 12, "completed": 2}
   - resigned (Boolean) - Optional field. Indicates if they left the company.

   a. Find employees hired exactly in the year 2020. Show only the name and dept fields, and hide the _id field.
   b. Find employees who have exactly both "Java" and "Python" included in their skills array.
   c. Find all employees whose name starts with the word "John".
   d. Find all employees where the number of projects.completed is exactly equal to the number of projects.assigned.
   e. Find the average salary of employees specifically working in the "IT" department.
   f. Flatten the skills array to find the most popular skill in the company. Count the total occurrences of each skill and sort them from highest to lowest.
7. Explain spark architecture in detail with a diagram. Marks: 4 + 2

   OR

   Explain about the ecosystem or unified stack of Apache Spark in detail. Why programs are lazily evaluated in Apache Spark. Marks: 4 + 2
8. Explain the architecture of HBase by depicting regions and region servers. Write three features of NoSQL over SQL. Marks: 3 + 3

   OR

   Explain fundamental differences between HBase and a traditional Relational Database Management System (RDBMS) in terms of schema, transactions, and data volume handling. Also write queries to create your own table, insert data and retrieve data to and from HBase. Marks: 3 + 3
9. Explain CAP theorem with an example. Marks: 6
10. Explain Hive architecture in detail with Hive services and Hive metastore. Differentiate between Apache Pig and Hive. Marks: 4 + 2

First Reassessment 2083

Subject: **Techniques for Big Data**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
PassMarks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Explain the 5 V's of Big Data with examples.
2. What is the purpose of ZooKeeper in the Hadoop architecture?
3. Explain the different Hadoop configuration modes.
4. Write short notes on YARN (Yet Another Resource Negotiator).
5. What is the use of the reduce() function in the MapReduce architecture? Explain with an example.

Group B [5x6=30]

6. Explain the distributed execution overview of MapReduce.
7. Explain the HDFS architecture along with its running daemons. Which daemon is responsible for storing all metadata of DataNodes, and why is it a single point of failure in Hadoop 1.x?

   OR

   Explain the anatomy of read and write operations in the Hadoop Distributed File System (HDFS).
8. Explain the core components of Hadoop along with all daemons. Also, explain Hadoop Streaming.

   OR

   Explain the anatomy of a MapReduce job run, including failures, shuffle, and sort phases, with reference to a word frequency count example.
9. Write HDFS commands with examples.
10. Why are Big Data technologies required? Explain the Big Data ecosystem in detail.